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C:\AyushSR\DAA\lab1.exe

Enter a number: 6
Factorial of 6 is 720
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C:\AyushSR\DAA\lab2.exe

Enter a number: 4
Factorial of 4 is 24
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C:\AyushSR\DAA\lab3.exe

Enter two numbers: 6 3

Tracing Table:
Iteration    A    B    R=A % B
1            6    3    0

GCD is 3
```

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C:\AyushSR\DAA\lab4.exe

Choose Method to Generate Fibonacci Series
1. Iteration
2. Recursion
Enter your choice: 1
Enter number of terms: 3

Tracing Table (Iteration):
Step  A    B    C
0     0    -    -
1     0    1    -
2     0    1    1

Fibonacci Series: 0 1 1
```

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C:\AyushSR\DAA\lab4.exe

Choose Method to Generate Fibonacci Series
1. Iteration
2. Recursion
Enter your choice: 2
Enter number of terms: 4

Tracing Table (Recursion):
i    fib(i)
0    0
1    1
2    1
3    2

Fibonacci Series: 0 1 1 2
```

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C:\AyushSR\DAA\lab6.exe

Enter number of elements: 3
Enter elements:
3 2 7
```

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Tracing Table (Bubble Sort):
Pass  Array  State
1     2 3 7
2     2 3 7
```

```
Tracing Table (Selection Sort):
Pass  Array  State
1     2 3 7
2     2 3 7
```

```
Tracing Table (Insertion Sort):
Pass  Array  State
1     2 3 7
2     2 3 7
```

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C:\AyushSR\DAA\lab5.exe

Enter number of elements: 4
Enter elements:
3 2 4 1
Enter element to search: 2

Tracing Table:
Step  Index  Element  Comparison
1     0      3      Not Matched
2     1      2      Match Found

Element found at position 2
Total comparisons = 2
```

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C:\AyushSR\DAA\lab7.exe

Enter number of elements: 3
Enter sorted elements:
1 3 4
Enter element to search: 4

Tracing Table (Iterative Binary Search):
Step  Low  High  Mid  Element  Result
1     0    2    1    3        Go Right
2     2    2    2    4        Found

Element found at position 3
```

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Tracing Table (Recursive Binary Search):
Step  Low  High  Mid  Element  Result
1     0    2    1    3        Go Right
2     2    2    2    4        Found

Element found at position 3
```

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C:\AyushSR\DAA\lab8.exe

Enter number of elements (even): 4
Enter elements:
3 2 5 6

Tracing Table
Low    High    Min    Max
-----
0      1      2      3
2      3      5      6
0      3      2      6

Final Result:
Minimum = 2
Maximum = 6

Total Comparisons = 4
Theoretical Comparisons = 4
```

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C:\AyushSR\DAA\lab9.exe

Choose Sorting Algorithm
1. Merge Sort
2. Quick Sort
Enter choice: 1

Enter number of elements: 3
Enter elements:
4 7 1

Merge Sort Tracing Table:
Merging [0...1] : 4 7
Merging [0...2] : 1 4 7

Sorted Array:
1 4 7
```

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C:\AyushSR\DAA\lab9.exe

Choose Sorting Algorithm
1. Merge Sort
2. Quick Sort
Enter choice: 2

Enter number of elements: 3
Enter elements:
6 2 8

Quick Sort Tracing Table:
Pivot = 8 | 6 2 8
Pivot = 2 | 2 6

Sorted Array:
2 6 8
```

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C:\AyushSR\DAA\lab10.exe
Enter number of elements: 4
Enter elements:
6 4 8 2

Initial Array:
6 4 8 2

Building Max Heap:
Heapify at index 0 : 8 4 6 2

Heap Sort Steps:
Swap root with index 3 : 2 4 6 8
Heapify at index 0 : 6 4 2
Swap root with index 2 : 2 4 6 8
Heapify at index 0 : 4 2
Swap root with index 1 : 2 4 6 8

Sorted Array:
2 4 6 8

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C:\AyushSR\DAA\lab11.exe
Greedy Algorithms
1. Fractional Knapsack
2. Job Sequencing with Deadlines
Enter choice: 1

Enter number of items: 3
Enter weight and profit of each item:
10 60
20 100
30 120
Enter knapsack capacity: 50

Tracing Table (Fractional Knapsack)
Wt    Profit  Ratio    Taken
-----
10     60      6        1
20    100      5        1
30    120      4       0.666667

Maximum Profit = 240

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C:\AyushSR\DAA\lab11.exe
Greedy Algorithms
1. Fractional Knapsack
2. Job Sequencing with Deadlines
Enter choice: 2

Enter number of jobs: 3
Enter job id, deadline, profit:
A 2 100
B 1 50
C 2 70

Tracing Table (Job Sequencing)
Job    Deadline    Profit    Slot
-----
A       2          100       2
C       2           70       1

Job Sequence: C A

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C:\AyushSR\DAA\lab12.exe
Enter number of vertices: 3
Enter number of edges: 3
Enter edges (u v weight):
0 1 2
1 2 3
0 2 1

Tracing Table (Kruskal's Algorithm):
Step    Edge(u,v)    Weight    Action
1       (0,2)         1        Selected
2       (0,1)         2        Selected
Total MST Cost (Kruskal) = 3

Tracing Table (Prim's Algorithm):
Step    Edge(u,v)    Weight
1       (0,2)         1
2       (0,1)         2
Total MST Cost (Prim) = 3

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C:\AyushSR\DAA\lab13.exe
Enter number of items: 3
Enter weights:
10 20 30
Enter profits:
60 100 120
Enter knapsack capacity: 50

Tracing Table (Greedy Method):
Item    Weight    Profit    Status
1       10        60        Selected
2       20       100        Selected
3       30       120        Rejected
Total Profit (Greedy) = 160

Tracing Table (DP Table):
Items\Cap    0    10    20    30    40    50
0             0     0     0     0     0     0
1             0    60    60    60    60    60
2             0    60   100   160   160   160
3             0    60   100   160   180   220
Total Profit (DP) = 220

Total Profit (Backtracking) = 220

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